Joo-Young Catherine Gonzalez

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Education

University of South Florida Tampa, FL
Master of Computer Science; GPA: 4.0 *August 2024 - May 2026*
Relevant Coursework: Operating Systems, Theory of Algorithms, Computer Architecture, Artificial Intelligence, Trustworthy Infrastructures, Social Media Mining, Affective Computing, Agentic AI, Trustworthy AI Systems

Technical Skills

Languages: Python, C, JavaScript, SQL
AI/ML: PyTorch, Hugging Face Transformers, LangGraph, spaCy, scikit-learn, LLM evaluation
Data & Visualization: pandas, NumPy, SciPy, Matplotlib
Systems & Tools: Linux, Bash, Git/GitHub, SLURM, Jupyter, Google Colab, Cursor

Projects

Multi-Agent Social Media Claim Verification
LangGraph, Python, Qwen 7B, Tavily, LLM Agents
- Designed and implemented the core LangGraph architecture for a two-person AI verification prototype that evaluates pasted social-media claims using pro/con debate agents, retrieved web evidence, an advisor, and a final verifier.
- Built the parent orchestration graph, debate agents, shared state, routing logic, and user interface, coordinating evidence and debate history across isolated subgraphs while preventing redundant searches.
- Integrated Qwen 7B with Tavily web-search evidence to generate supported, refuted, or insufficient-evidence verdicts with written rationales; used schema-validated JSON outputs and a heuristic fallback to handle malformed model responses.

Text-to-Image Generative AI + Robustness Evaluation
PyTorch, DCGAN, Stable Diffusion, CLIP, BLIP VQA
- Contributed architecture research and experimental planning for a three-person project that built a text-conditioned DCGAN in PyTorch using MS COCO captions, GloVe embeddings, and 64x64 image generation.
- Analyzed training instability, poor image quality, and weak text-image alignment to document limitations of the generated outputs and model design.
- Designed and implemented a robustness benchmark for Stable Diffusion models using 100 logically equivalent prompt pairs across negation, word order, composition, attributes, and spatial relations, evaluated with CLIP-based alignment and BLIP VQA.
- Identified recurring failures in negation, multi-object composition, and spatial reasoning, including substantial inconsistency under simple meaning-preserving prompt reordering.

Dissent in Reddit Discussions — Social Media Mining
Python, PyTorch, Hugging Face Transformers, RoBERTa, NLP
- Proposed the core project direction and conducted the background research for a team study examining how dissent relates to engagement and conversational uptake across Reddit communities.
- Developed a RoBERTa-based stance-classification pipeline using DEBAGREEMENT parent–reply pairs, achieving 0.678 weighted F1 on agreement, neutral, and disagreement labels.
- Built and debugged a scalable preprocessing and batch-inference pipeline for 2.43 million comment–reply pairs across 35,569 threads, reorganizing noisy source files by thread ID and using seeded whole-thread sampling to preserve complete conversations for downstream analysis.

Experience

Research Assistant — TKAI Lab
Tampa, FL
AI & NLP Research *Oct. 2025 – May 2026*
- Developed a corpus-analysis pipeline in Python and spaCy that processed ICNALE learner essays across A2–B2 bands and extracted POS distributions, dependency relations, morphological categories, sentence-length measures, and clause-marker frequencies.
- Synthesized research across second-language acquisition, CEFR validity, criterial features, linguistic complexity, automated essay assessment, knowledge tracing, and LLM evaluation to reframe proficiency as a multidimensional learner-state modeling problem.
- Developed and revised research proposals under faculty guidance, helping translate theoretical findings into plans for interpretable, multidimensional learner modeling using linguistic and neural representations.

Research Assistant — CEREAL Lab
Tampa, FL
Evolutionary Algorithms *Jan. 2024 – May 2024*
- Conducted Python-based experiments using genetic algorithms and other evolution-inspired metaheuristic techniques to solve Boolean SAT instances and evaluate performance under different parameter settings.
- Managed experiment runs on a computing cluster using Linux CLI, SLURM, and Git to support job scheduling, execution, and version control.